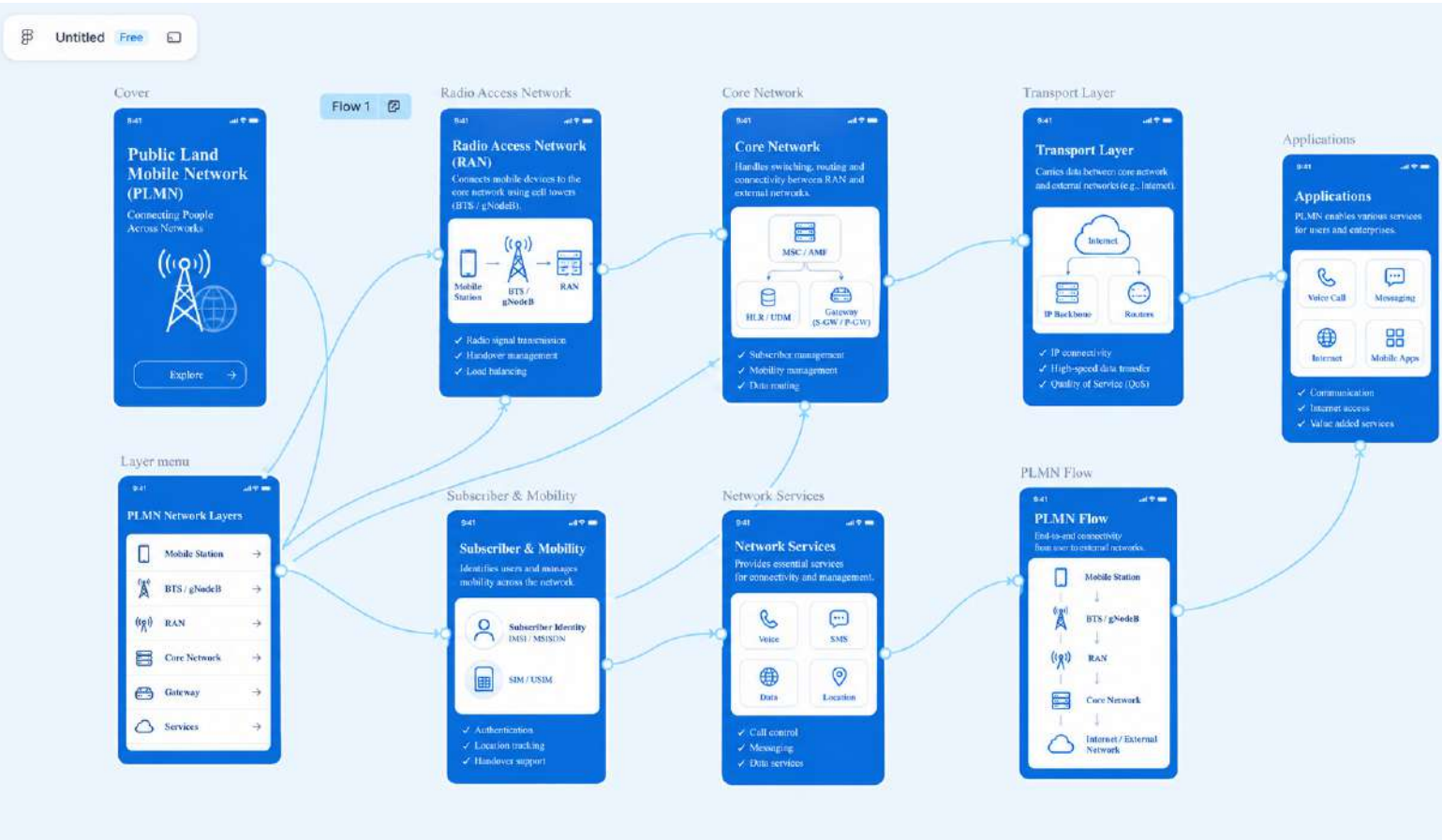

Untitled
Free
Cover
Public Land Mobile Network (PLMN)
Connecting People Across Networks
Explore
Flow 1
Radio Access Network
Radio Access Network (RAN)
Connects mobile devices to the core network using cell towers (BTS / gNodeB).
Mobile Station
BTS / gNodeB
RAN
Radio signal transmission
Handover management
Load balancing
Core Network
Core Network
Handles switching, routing and connectivity between RAN and external networks.
MSC / AMF
HLR / UDM
Gateway (S-GW / P-GW)
Subscriber management
Mobility management
Data routing
Transport Layer
Transport Layer
Carries data between core network and external networks (e.g., Internet).
Internet
IP Backbone
Routers
IP connectivity
High-speed data transfer
Quality of Service (QoS)
Applications
Applications
PLMN enables various services for users and enterprises.
Voice Call
Messaging
Internet
Mobile Apps
Communication
Internet access
Value added services
Layer menu
PLMN Network Layers
Mobile Station
BTS / gNodeB
RAN
Core Network
Gateway
Services
Subscriber & Mobility
Subscriber & Mobility
Identifies users and manages mobility across the network.
Subscriber Identity
IMSI / MSISDN
SIM / USIM
Authentication
Location tracking
Handover support
Network Services
Network Services
Provides essential services for connectivity and management.
Voice
SMS
Data
Location
Call control
Messaging
Data services
PLMN Flow
PLMN Flow
End-to-end connectivity from user to external networks.
Mobile Station
BTS / gNodeB
RAN
Core Network
Internet / External Network